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Code: 2503/R

Faculty of Science

BCA I-Year, CBCS-II Semester Regular Examinations June-2024

PAPER: COMPUTER ARCHITECTURE

Time: 3 Hours **Max Marks: 70**

Section-A

I. Answer any FIVE of the following questions **(5x4=20 Marks)**

1. Write about Bus Structures
2. Explain addition of signed numbers
3. Write about Data hazards?
4. Write the difference between RAM and ROM?
5. Write about Interrupts?
6. Explain about virtual memory?
7. Write about software performance
8. Write about Semiconductors

Section-B

II. Answer the following questions **(5x10=50 Marks)**

9. (a) Explain about Memory operations?

OR

(b) Write about Basic I/O operations?

10. (a) Explain about Multiplication of positive numbers with example ?

OR

(b) explain about Floating point numbers and operations ?

11. (a) Write about Microprogrammed control?

OR

(b) Write about Data path and control considerations ?

12. (a) Explain about Memory management requirements?

OR

(b) Write about Secondary storage ?

13. (a) Explain about Accessing I/O devices?

OR

(b) Write about Interface circuits ?

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Code: 2203/R

Faculty of Science

BCA I-Year, CBCS-II Semester Regular Examinations June-2024

PAPER: OBJECT ORIENTED PROGRAMMING WITH C++

Time: 3 Hours

Max Marks: 70

Section-A

I. Answer any FIVE of the following questions **(5x4=20 Marks)**

1. What are the Applications of OOP?
2. What is goto statement?
3. What is friend function?
4. What is multiple inheritance?
5. Define Virtual Function.
6. What are Basic concepts of OOP?
7. Define Constructor.
8. What is Template?

Section-B

II. Answer the following questions **(5x10=50 Marks)**

9. (a) Explain about storage Classes in C++ with example

OR

(b) Explain about operators used in C++ with Example.

10. (a) Explain about if, else if and else if ladder statements with example.

OR

(b) What is Function? Explain about Call by Reference with C++ program.

11. (a) Explain about C++ Constructors and types with examples.

OR

(b) What is Friend Function? Write C++ program on Friend Function.

12. (a) Define Inheritance and Explain its Types with Examples.

OR

(b) What is Pointer? Explain about Pointer arithmetic with example.

13. (a) What is Exception? Explain about C++ exception handling.

OR

(b) Explain about Polymorphism and its types with Examples.

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Code: 2504/R

Faculty of Science

BCA I-Year, CBCS-II Semester Regular Examinations June-2024

PAPER: DATA STRUCTURES

Time: 3 Hours **Max Marks: 70**

Section-A

I. Answer any FIVE of the following questions **(5x4=20 Marks)**

1. Define Array and Explain about types of Arrays?
2. Define Stack and operations of stack?
3. Write about Hash functions?
4. Write about Binary Tree Traversal?
5. Define Sorting and explain about Bubble sort?
6. Define Graph and write about terminology?
7. Write about Binary Search ?
8. Write about Abstract Data Types ?

Section-B

II. Answer the following questions **(5x10=50 Marks)**

9. (a) Define class and explain with example program in c++?

OR

(b) Explain about matrices and explain about special matrices and sparse matrices?

10. (a) Explain about Queue and operations with example program?

OR

(b) Explain about stack operations with example program ?

11. (a) Define Linked List and types of Linked Lists?

OR

(b) Define Hashing and explain about static and Hash Tables?

12. (a) Define Tree and explain about Binary Tree and its operations?

OR

(b) Write about DFS with example?

13. (a) Explain about Selection sort with example program?

OR

(b) Write about binary search with example program?

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Code: 2505/R

Faculty of Science

BCA I-Year, CBCS-II Semester Regular Examinations June-2024

PAPER: Advance Computer Networks

Time: 3 Hours **Max Marks: 70**

Section-A

I. Answer any FIVE of the following questions **(5x4=20 Marks)**

1. Explain about high speed Networks?
2. Explain about advantages and disadvantages of ATM?
3. Explain about IPv4 address space?
4. Write the differences between Intradomain and Interdomain routing?
5. What is network traffic engineering?
6. Explain about Overlay networks?
7. Explain about advantages and disadvantages of IPv6?
8. What is Domain Name System?

Section-B

II. Answer the following questions **(5x10=50 Marks)**

9. (a) Explain about Network Architecture?

OR

(b) Explain the CRC technique with an example?

10. (a) what is Datagram Network? Explain about features of Datagram Network?

OR

(b) What is ISDN? Explain about ISDN Services.

11. (a) Explain about classful IP Addressing in detail?

OR

(b) Explain about Congestion control algorithms?

12. (a) What is Congestion Control? Explain about Congestion-Avoidance Mechanisms?

OR

(b) Explain about Major Protocols of Unicast Routing in detail?

13. (a) Explain about SONET/SDH Transmission with a neat diagram ?

OR

(b) Explain about Pretty Good Privacy (PGP) protocols and Firewalls.?

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Code: 2501/R

Faculty of Science

BCA I-Year, CBCS-II Semester Regular Examinations June-2024

PAPER: FUNDAMENTALAS OF PROBABILITY& STATISTICS

Time: 3 Hours | Max Marks: 70

Section-A

I. Answer any FIVE of the following questions (5x4=20Marks)

1. Define collection of methods.
2. What are the measures Central tendency, explain any two of them.
3. Define the definition of Mathematical probability.
4. Explain Probability mass function and Probability density function.
5. Define Correlation.
6. Define Classification of data.
7. Define Skewness.
8. Define Regression.

Section-B

II. Answer the following questions (5x10=50Marks)

9. (a) Explain briefly about Classification and Tabulation of data.

OR

(b) Explain concept of Statistics population, Sample and Quantitative data.

10. (a) What are the Measures of Dispersion. Explain briefly.

OR

(b) Define Skewness and Kurtosis with it's formula.

11. (a) Define Addition theorem of probability for 2 events.

OR

(b) Define Multiplication theorem of probability for 2 events.

12. (a) Explain the Expectation of Random variable and mention it's probabilities.

OR

(b) Define Binomial theorem.

13. (a) Define Partial and Multiple Correlations with it's formulae.

OR

(b) Explain test procedure of 'Paid t-test'.